

RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT

Tutorial 12 Temperature and Ideal Gases Suggested Solutions

- D1 (a)** No. of mol of iron = $\frac{m}{M} = \frac{6.3 \times 10^{-3} \text{ kg}}{5.6 \times 10^{-2} \text{ kg mol}^{-1}} = 0.1125 \text{ mol}$
No. of iron atoms = No. of mol $\times N_A = 0.1125 \times 6.02 \times 10^{23} = 6.8 \times 10^{22}$
- (b)** Maximum volume = $\frac{\text{Total volume}}{\text{No. of atoms}} = \frac{(20 \times 10^{-3})^3}{6.8 \times 10^{22}} = 1.2 \times 10^{-28} \text{ m}^3$
- (c)** Volume = $d^3 \Rightarrow d = \sqrt[3]{\text{Volume}} = \sqrt[3]{1.2 \times 10^{-28}} = 4.9 \times 10^{-10} \text{ m}$

The atoms are assumed to be hard spheres, each occupying a cube whose sides are equal to the diameter of an atom.

Note: it is incorrect to determine diameter d based on volume of a sphere, i.e.

$$V_{\text{sphere}} = \frac{4}{3} \pi \left(\frac{d}{2} \right)^3 = 1.2 \times 10^{-28} \text{ m}^3 \Rightarrow d = 6.1 \times 10^{-10} \text{ m} \text{ is wrong.}$$

- D2 (a)** $pV = nRT$
$$n = \frac{pV}{RT} = \frac{(1.0 \times 10^5)(0.064)}{(8.31)(27 + 273.15)} = 2.566 \text{ mol}$$

$$\text{Mass of air} = \text{no. of moles} \times \text{molar mass} = (2.566)(0.030) = 0.077 \text{ kg}$$

- (b)** $pV = n'RT'$
$$n' = \frac{pV}{RT'} = \frac{(1.0 \times 10^5)(0.064)}{(8.31)(180 + 273.15)} = 1.700 \text{ mol}$$

$$\Delta n = 2.566 - 1.700 = 0.866 \text{ mol}$$

$$\Delta m = (0.866)(0.030) = 0.026 \text{ kg}$$

D3 At equilibrium position, the pressures on the two sides of the piston are equal.

For an ideal gas, $pV = nRT$.

Initially, $n \propto V$ since P and T are constants,

$$\frac{n_A}{n_B} = \frac{V_A}{V_B} = \frac{2}{3}$$

After temp in A is raised and equilibrium is achieved, $V \propto nT$ since p is a constant,

$$\frac{V_A'}{V_B'} = \frac{n_A T_A'}{n_B T_B'} = \frac{2}{3} \times \frac{177 + 273.15}{27 + 273.15} = 1.0$$

The piston is now at the middle of the cylinder (because now $V_A = V_B$).

So, the piston has moved $0.25 - 0.20 = 0.05$ m to the right.

D4 (a) $pV = NkT$

$$pV = N \left(\frac{2}{3} \right) \left(\frac{3}{2} kT \right)$$

$$N = \frac{pV}{\left(\frac{2}{3} \right) \langle E_k \rangle} = \frac{(1.20 \times 10^5)(4000 \times 10^{-6})}{\left(\frac{2}{3} \right) (3.60 \times 10^{-22})} = 2.00 \times 10^{24} \text{ atoms}$$

$$n = \frac{N}{N_A} = \frac{2.00 \times 10^{24}}{6.02 \times 10^{23}} = 3.32 \text{ mol}$$

(b) $E_{k, \text{gas}} = \frac{3}{2} kT \times N_A = \frac{3}{2} RT = \frac{3}{2} (8.31)(25 + 273.15) = 3.72 \times 10^3 \text{ J}$

$$E_{k, \text{sprinter}} = \frac{1}{2} mv^2 = \frac{1}{2} (70) \left(\frac{100}{10} \right)^2 = 3.50 \times 10^3 \text{ J}$$

$$\therefore \frac{\langle E_k \rangle_{\text{gas}}}{\langle E_k \rangle_{\text{sprinter}}} = 1.1$$

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YEAR 56 PHYSICS DEPARTMENT

- D5**
- For each elastic collision between a molecule and a wall, the change in momentum of the molecule is proportional to its speed: $\Delta P = -2mc_x$
 - The time interval for this molecule to collide with the same wall again is inversely proportional to the speed: $\Delta t = \frac{2d}{c_x}$
 - The average force exerted on the molecule by the wall is given by Newton's second law: $F_{\text{on molecule}} = \frac{\Delta P}{\Delta t} = \frac{-2mc_x}{2d/c_x} = -\frac{mc_x^2}{d}$
 - The average force exerted on the wall is given by Newton's third law: $F_{\text{on wall}} = -F_{\text{on molecule}} = \frac{mc_x^2}{d}$, which is proportional to speed squared.
 - When adding up the forces due to each of the molecules, the total force is proportional to the sum of all speeds squared, which is just the mean-square speed multiplied by the number of particles.
 - Since pressure is just force divided by the area, pressure is also proportional to the mean square speed. (Refer to pg 11-12 of the notes.)

- D6 (a)**
- $$pV = NkT \Rightarrow p = \frac{N}{V}kT$$
- Thus, $n = \frac{N}{V}$ is the number density (number of particles per unit volume) and k is the Boltzmann constant.

(b)

$$pV = \frac{1}{3}Nm\langle c^2 \rangle \Rightarrow p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

Since Nm is the total mass of the gas, $\rho_m = \frac{Nm}{V}$ is the mass per unit volume, i.e. the density of the gas.

- D7 (a)**
- From $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$, $c_{\text{rms}} = \sqrt{\frac{3kT}{m}} \Rightarrow c_{\text{rms}} \propto \sqrt{\frac{1}{m}}$ for a given temperature

$$\frac{(c_{\text{rms}})_p}{(c_{\text{rms}})_e} = \sqrt{\frac{m_e}{m_p}}$$

$$(c_{\text{rms}})_p = \sqrt{\frac{m_e}{m_p}} \times (c_{\text{rms}})_e = \sqrt{\frac{9.11 \times 10^{-31}}{1.67 \times 10^{-27}}} \times 3.0 \times 10^6 = 7.0 \times 10^4 \text{ m s}^{-1}$$

(b)

$$c_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

$$\begin{aligned}
 T &= \frac{m_e (c_{rms})_e^2}{3k} \\
 &= \frac{9.11 \times 10^{-31}}{3 \times 1.38 \times 10^{-23}} (3.0 \times 10^6)^2 \\
 &= 2.0 \times 10^5 \text{ K}
 \end{aligned}$$

- D8**
- (a) $pV = nRT$
 $V = \frac{nRT}{P} = \frac{(1)(8.31)(273)}{1.01 \times 10^5} = 2.25 \times 10^{-2} \text{ m}^3$
- (b) $\rho = \frac{m}{V} = \frac{40 \times 10^{-3}}{2.25 \times 10^{-2}} = 1.78 \text{ kg m}^{-3}$
- (c) Using $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
 $c_{rms} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.01 \times 10^5)}{1.78}} = 413 \text{ ms}^{-1}$
- (d) $\langle E_k \rangle = \frac{3}{2} kT = \frac{3}{2} (1.38 \times 10^{-23})(273) = 5.65 \times 10^{-21} \text{ J}$

- D9**
- (a) Fast molecules can move at speeds greater than the escape velocity of the Earth, thus they are able to escape the atmosphere to infinity.
- (b) At thermal equilibrium, the average microscopic kinetic energies of all molecules are the same. Thus, lighter molecules generally travel faster (since speed is inversely proportional to the square-root of mass) and are able to escape from the Earth.
 Massive molecules can only move at lower speeds that are smaller than the Earth's escape velocity and hence they are unable to escape.
- (c) For an ideal gas, the root-mean-square speed of the gas molecules is directly proportional to the square-root of temperature. At any temperature, there will be a small portion of the gas molecules that move at speeds much higher than the root-mean-square speed. These are the particles who could possess speeds greater than the escape speed and hence are able to escape.
 (At that same temperature, there will also be a small portion of the molecules who possess speeds much lower than the root-mean-square speed.)
- (d) It is more energetically favourable for hydrogen to combine with available oxygen to form the compound water than to remain as molecular hydrogen.
 Intermolecular forces between water molecules are very strong as a result of hydrogen bonding. Thus, water has a very high boiling point (much higher than Earth's surface temperatures) such that water exist mainly in its liquid state on the surface of the Earth and will be unable to escape.
 In its gaseous state (i.e. water vapour), these molecules are too heavy (compared to H_2) to possess sufficient speeds to escape Earth. Furthermore, low temperatures in the Earth's atmosphere will cause these water molecules to condense as clouds and fall back to Earth as rain or snow.

RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT

Given that a large fraction of the hydrogen molecules possess sufficient speeds to escape Earth, large quantities of hydrogen can thus only be found as a constituent of water which remains trapped on Earth.

- C1 (a)** Assume that air behaves like an ideal gas,

$$\text{ave KE of a molecule} = \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

$$\langle c \rangle = \sqrt{\frac{3kT}{m}}$$

Ideal gas equation,

$$pV = NkT$$

$$\text{no. density, } n = \frac{N}{V} = \frac{p}{kT}$$

$$S = \frac{1}{4} n c A = \frac{1}{4} \frac{p}{kT} \left(\frac{3kT}{m} \right)^{1/2} A = \frac{\sqrt{3}}{4} \frac{p}{\sqrt{mkT}} A$$

Since air pressure outside the tube is higher than the gas pressure inside the tube, air molecules will leak into the tube from the outside.

Rate at which air molecules leak into the tube,

$$S = \frac{\sqrt{3}}{4} \frac{p}{\sqrt{mkT}} A$$

$$= \frac{\sqrt{3}}{4} \frac{1.0 \times 10^5}{(5.6 \times 10^{-26} \times 1.38 \times 10^{-23} \times 300)^{1/2}} (8.0 \times 10^{-17})$$

$$= 2.28 \times 10^{11} \text{ molecules s}^{-1}$$

(Assume that the temperature inside and outside the tube is the same.)

- (b)** Since the pressure in the tube is much smaller than the pressure outside the tube, it may be assumed that the rate at which the gas molecules leave the tube is negligible compared to the rate at which air molecules enter the tube. The increase in pressure in the tube is mainly attributed to the increase in the number of air molecules in the tube. The rate at which air molecules enter the tube may be assumed approximately constant.

Ideal gas equation,

$$pV = NkT$$

$$\text{No. of molecules in the tube, } N = \frac{pV}{kT}$$

No. of air molecules that entered tube to increase the pressure in the tube, $N_2 - N_1$

$$= \frac{V}{kT} (p_2 - p_1)$$

$$= \frac{2.0 \times 10^{-3}}{(1.38 \times 10^{-23})(300)} (5.0 \times 10^{-3} - 1.0 \times 10^{-6})$$

$$= 2.415 \times 10^{15} \text{ molecules}$$

$$\text{Time taken, } t = \frac{N_2 - N_1}{S} = \frac{2.415 \times 10^{15}}{2.275 \times 10^{11}} = 10615 \text{ s} = 177 \text{ min} = 2.95 \text{ h}$$